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ALA - Assignment - 1

Q1

a) Distinguish between a linearly independent and a linearly dependent set of vectors. Analyse why a set of 2 non-zero vectors $\{u, v\}$ is linearly dependent exactly when one of them is multiple of the other.

let set of vectors: $v_1, v \dots V_K$

Consider the equation $C_1v_1 + C_2v_2 + \dots C_K v_K = 0$

① the set of vectors is said to be linearly independent if the eqⁿ has only trivial solⁿ like $c_1 = c_2 = \dots = c_k = 0$

② the set of vector is said to be linearly dependent if there exists a non-trivial solⁿ to the eqⁿ.
where atleast 1 coefficients non-zero.

To prove: if 2 non-zero vectors are dependent, one is a multiple of the other.

↳ the vectors are linearly dependent: $au + bw = 0$ for some constants a, b.
$\{u, w\}$

the eqⁿ has non trivial solⁿ

rearranges ⇒ $au = -bw$

since u & w are non-zero, & a non trivial solⁿ requires $a \neq 0$, but a.

$\therefore \Rightarrow u = -\frac{b}{a} w$.

↳ this implies u is a scalar multiple of w.

& inversely

suppose $w = xu$.

rearrange is $xu - w = 0$

coefficient = $x, -1$ ∴ non trivial linear combination produces zero vector.

Hence, vectors are linearly dependent.

∴ Hence proved, a set of 2 non-zero vectors is linearly dependent iff one vector is scalar multiple of another

b) Let $v_1 = (1,0,2)^T$, $v_2 = (2,1,3)^T$ & $v_3 = (0,1,-1)^T$. Examine whether v_3 can be written as $av_1 + bv_2$ by determining the values of a & b. Deduce whether the set $\{v_1, v_2, v_3\}$ is linearly independent, justify.

→ Goal:

decide whether $\{v_1, v_2, v_3\}$ is linearly independent.

Soln :-

A vector a is a linear combination of other vectors if it can be expressed as: $v_3 = av_1 + bv_2$

a set of vectors is linearly dependent if a non trivial linear combination produces the zero vector.

Step ① - Assume linear combination $v_3 = av_1 + bv_2$

Substitute given vectors

$$\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = a\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + b\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

↳
$$0 = a(1) + b(2) \quad \text{—①}$$
$$1 = a(0) + b(1) \quad \text{—②}$$
$$-1 = 2a + 3b \quad \text{—③}$$

② → $b = 1$

then ① & b gives

$$0 = a + (b)(2) = a = -2$$

putting a, b in 3rd

→
$$2(-2) + 3(1) = -1$$
$$-4 + 3 = -1 = -1$$ ← which is true

hence $-v_3 = 2v_1 + 3v_2$

answer = $2v_1 + 3v_2 + v_3 = 0$

coeff = $(2, 3, 1)$ are not all zeros.

∴ so a non-trivial combination gives 0 ∴ set is linearly dependent.